

LEGAEDIT: A Dissociation Diagnostic for Legal Meaning Preservation Metrics

Gaurab Baral

University of Cincinnati

baralgb@mail.uc.edu

Abstract

Does a simplified legal clause still say what the original said? The checks in current use cannot establish that it does: requiring an identical pair to score highest and an unrelated pair lowest moves lexical overlap and legal force together, so any monotone function of token overlap satisfies both. Our remedy is a *dissociation*, an item holding surface form fixed while legal force moves. We release LEGAEDIT, 373 minimal perturbations of Quebec statutory French that reverse legal force while preserving 0.93 of the tokens. Each of the seven embedding and BERTScore metrics we test spends only 0.022–0.039 of its identical-to-unrelated range on such an edit, against 0.67 for bidirectional NLI, the one family the identical-pair check would disqualify. On FRJUDGE the two orderings disagree at both extremes: the best-correlating signal is a length feature with the lowest margin we measure, and the family that moves on the edit sits mid-table on correlation.

1 Introduction

Rewriting a legal text for a lay reader has an unusual failure mode. When a news summary drops a qualifier the reader is mildly misinformed; when an insurance clause drops a qualifier the reader may be uninsured. The 2024 decision in *Moffatt v. Air Canada*, which held an airline liable for its chatbot’s misstatement of its own bereavement-fare policy, established that the gap between what a document says and what an automated system says it says is legally cognisable and not merely academic.

Nor is the scale hypothetical any longer. Hariri and Ho [27] document a U.S. state claiming to have eliminated a third of its code with automated assistance, and measure what such systems actually deliver: an F1 of 0.67 on a benchmark built from the U.S. Department of Labor’s own six-month manual reconciliation of unemployment insurance law across fifty states. On the consumption side,

Magesh et al. [38] find that the commercial legal research tools marketed as having eliminated hallucination still produce it on 17 to 33% of queries. Automated rewriting of law is being deployed at statutory volume. What is validated is an aggregate, and what is at stake is one clause at a time.

If simplified legal text is produced automatically, something has to check it, and that something will in practice be an automatic metric. What a practitioner needs to know of it is narrower and harder than correlation with human ratings on naturally occurring simplifications: does the score move when the legal effect of the clause moves, and does it stay put when the wording changes without legal effect? Neither is measured by a correlation coefficient. Worse, the validation apparatus the field has converged on cannot, even in principle, distinguish a metric that reads law from a metric that counts shared words.

Two automated checks are in current use [7, 8]: a sentence paired with itself must receive the maximum score, and a sentence paired with an unrelated one the minimum. Both are necessary and both are easy to state. But moving from an identical pair to an unrelated pair changes lexical overlap and legal meaning *together*, in the same direction, at the same time. Any monotone function of token overlap therefore satisfies both checks by construction, which is exactly why the embedding metrics in the literature already pass them. The checks test calibration at two endpoints; they do not test what the metric is reading.

The remedy is a *dissociation*: an evaluation item on which the two confounded variables are pulled apart, so that legal force changes while surface form does not. Consider a clause in which *doit* (must) is replaced by *peut* (may). Ninety-odd percent of the tokens are untouched, the sentence remains fluent and statutory in register, and an obligation has become a permission. A metric that measures legal meaning must drop sharply;

a metric that measures overlap cannot. We build LEGAEDIT, 373 such perturbations over Quebec statutory French, and release it with a harness taking an arbitrary scoring function, so that untrained metrics, fine-tuned regressors and prompted judges can all be administered the same test.

LEGAEDIT separates the metric landscape cleanly (Section 5). The similarity metrics collapse, spending 0.022–0.039 of their working range on the edit; bidirectional NLI entailment spends 0.67. The diagnostic is therefore hard but not impossible, which makes a low score a fixable deficiency rather than an inherent limit of automatic evaluation. Awkwardly for current practice, the family that clears it is the one the identical-pair check would disqualify: multilingual NLI scores a sentence paired with itself at 0.780 rather than 1.

A diagnostic alone does not discipline how corpus-correlation evidence is read, so we assemble three further requirements around it (Section 3). The first is a *measured* human ceiling, which is free to compute whenever a corpus ships several ratings per item. The second is comparison under *identical supervision*, since an unsupervised similarity score and a regressor fitted to the label are not otherwise on the same scale. The third is a *trivial-feature control*, because the surface statistics of a simplification, above all how much shorter it is, carry more of the human label than is usually assumed.

We instantiate all four on FRJUDGE [7], 297 Quebec insurance clauses with five expert ratings each and the only annotated corpus for this construct (Section 6). It supports a measurable human ceiling of $r = 0.597$. A ridge regression over ten surface features reaches $r = 0.62$ on the same split, at that ceiling and above every semantic metric we test; the bare length difference does better still. And the rank correlation between corpus performance and LEGAEDIT margin is $\rho = 0.24$ over the 13 scorers we place on both axes, with a 95% bootstrap interval of $[-0.39, 0.75]$: at this many metrics the relationship cannot be resolved in either direction, while the two orderings disagree outright at both extremes.

Contributions.

- LEGAEDIT, a 373-item dissociation diagnostic defined by legal effect rather than by linguistic category, with a scoring harness and the margin statistic it is read with.
- An argument that the sanity checks in current use are jointly satisfiable by lexical overlap,

and a four-part protocol that repairs this.

- An empirical separation of the field into a similarity family that fails the diagnostic and an entailment family that largely clears it.
- The first human-ceiling estimate for legal meaning preservation, with a calibrated baseline comparison that includes trivial surface features.
- A prompted-judge baseline run through the same harness, with an independent replication and a prompt-sensitivity analysis, and the finding that the deduct-per-error rubric the corpus is annotated under is itself what limits the judge’s diagnostic sensitivity.

2 Related Work

Evaluating text simplification. Simplification quality is conventionally decomposed into fluency, simplicity and meaning preservation, and metrics borrowed from translation and summarisation transfer poorly to the third. Sulem et al. [50] show BLEU [42] does not correlate with meaning preservation once sentence splitting is involved, the operation most characteristic of legal simplification [22]; SARI [54] targets simplicity rather than adequacy, against general-domain references [2]. Embedding metrics [47, 61] capture semantic variability better but, as Cripwell et al. [15] and Kew et al. [30] document, score consistently high across *all* alterations, penalising even substantial meaning changes too little. The confound sits below the metric layer: on the VISLA benchmark [19], sentence encoders struggle to separate lexical from semantic variation at the representation level, so a metric built on those representations inherits the problem rather than causing it. Trained metrics such as LENS [37], MeaningBERT [8] and, most recently, DETECT for German [32] improve correlation on the corpora they are fitted to, and the pattern is now general rather than a one-off: a language without a trained metric acquires one. Agrawal and Carpuat [1] argue for grounding meaning preservation in reading comprehension rather than similarity, and Trienes et al. [51] characterise simplification errors as recoverable *information loss*, close to the omission category that dominates legal error taxonomies. The pressure on this literature has since increased from the system side. Qiang et al. [45] benchmark LLMs against supervised methods across lexical, syntactic, sentence and document simplification and find that the outputs frequently

exceed the quality of the human-annotated references themselves, which makes the metric, not the system, the binding constraint on progress.

French and legal resources. French simplification corpora are few and none are legal: Alector, CLEAR [23] and WikiLarge-FR [12] cover literacy, medical and encyclopaedic text, and the position has not changed. The most recent French simplification study we know of [16] evaluates seven LLMs and five prompting strategies over five corpora, all of them Wikipedia or medical. Legal NLP resources are mostly English and mostly not about simplification [28]; the current survey of legal LLM evaluation [29] organises the field around outcome correctness, reasoning reliability and trustworthiness, none of which is meaning preservation under rewriting. Hagan [26] proposes criteria a legal AI system should be benchmarked against. Two of them, that a response covers details and exceptions and that it does not misrepresent the substantive law, form the basis of the construct Beauchemin et al. [7] operationalise in FRJUDGE. That corpus is our case study, and to our knowledge it is still the only annotated resource for legal meaning preservation in any language.

Challenge sets and entailment. Our diagnostic follows work showing that correlation on naturally occurring outputs hides brittleness on targeted perturbations. Sai et al. [49] build perturbation checklists for NLG metrics and Anshütz et al. [3] show that metrics systematically miss negation. Closest to us, Chen and Eger [13] demonstrate that similarity-based metrics are not robust to meaning-changing edits while NLI-based metrics repair much of the gap. The complementary experiment has since been run: Mujahid et al. [39] hold meaning fixed and vary form, applying seven factuality-preserving perturbations to long-document summaries, and find that six widely used reference-free factuality metrics return inconsistent scores for semantically equivalent text. Taking the two together gives the shape of the problem, since a usable metric must move on the first kind of edit and stay put on the second, and current metrics reliably do neither. LEGALEDIT adapts the method to a domain where the perturbations are not linguistic curiosities but the exact operations that decide coverage disputes, and where the inventory is therefore organised by legal effect rather than by linguistic category; to our knowledge no prior challenge set targets legal force. The entailment side

of that literature is well developed for English adequacy [34, 59], which is why it is absent from the French literature. We construct a French analogue by running a multilingual NLI model [14, 35] in both directions, so that failure of $o \Rightarrow s$ signals hallucination and failure of $s \Rightarrow o$ signals omission. These are the two dominant categories of the legal error taxonomy, which is why we treat this family as the strongest available starting point rather than as one baseline among many. Starting point is the operative phrase. On biomedical plain language summarisation You and Guo [57] show entailment and QA-based consistency metrics breaking down because simplification for a lay reader legitimately adds explanatory content the source does not contain, which a strict entailment reading scores as unfaithful; we return to what that implies here in Section 7.

Whether a benchmark measures what it claims.

The objection this paper raises against two specific checks is an instance of a problem the field has begun to name. Reviewing 445 LLM benchmarks with 29 expert reviewers, Bean et al. [6] find recurring choices of phenomenon, task and scoring metric that undermine the validity of the claims those benchmarks are used to support, and ask authors to state what construct a score is evidence about before treating a difference in it as evidence of anything. Pacchiardi et al. [41] give the concrete version for internal validity: simple n -gram classifiers, possessing none of the capacity under test, score well on modern benchmarks, so a benchmark can be solved by a route that bypasses the construct entirely. Requirement 3 below is that control transposed from a capability benchmark to a metric comparison. Pita [44] makes the same point about instruments rather than systems, that a model agreeing with human coders is thereby reliable and not thereby valid, since it may reach the code through a correlate. The pattern recurs wherever the construct is not directly observable: Gur-Arieh et al. [25] construct ground-truth labels for chain-of-thought faithfulness and find that most published faithfulness metrics perform near chance, having been validated against proxies rather than against the property claimed. For automatic evaluation specifically, Deviyani and Diaz [18] argue that metric quality is not a global property at all, and that both the absolute and the relative standing of metrics shift with the evaluation context, which is our reason for reporting the diagnostic and the

corpus correlation as two answers rather than one ranking.

Disagreement and meta-evaluation. The assumption that a single gold label exists has been extensively challenged [43, 52], and Xu and Jurgens [55] survey the resulting shift from consensus learning toward modelling the disagreement itself. Legal interpretation is a paradigm case, since disagreement between competent lawyers is the ordinary condition of the field rather than annotator error, and we accordingly treat the five ratings per item as a distribution to be modelled (Section 6.6). Two recent results bear on how that distribution should be handled. Zhang and Çöltekin [60] study *graded* ratings specifically, which is the case here, and propose an opposition index for items whose annotators hold opposed rather than merely noisy positions; such items are harder to predict and are often underestimated by models. Bonagiri et al. [10] show that collapsing annotators by majority or mean discards item ambiguity and rater reliability together, and that system rankings built on the collapsed label are unstable across annotator subsets. Both describe directly what Section 6.1 finds in FR-JUDGE. On the statistical side, Deutsch et al. [17] show correlation estimates for generation metrics carry wide confidence intervals and that apparent metric differences frequently fail significance testing, Graham and Baldwin [24] give the appropriate test for dependent correlations, and Card et al. [11] show NLP comparisons are routinely underpowered. Our protocol applies all three as a matter of course.

3 What Would Validate a Legal Meaning Metric?

We first set out what we think the evidence should consist of. The four requirements below are stated for an arbitrary scorer $m(o, s) \in \mathbb{R}$ taking an original clause o and a candidate simplification s ; none presuppose an architecture, and all four are cheap enough that we see no reason for a metric paper to omit them. They are an attempt to answer, for this one construct, the question Bean et al. [6] put to benchmark authors in general: what would have to be true for this number to be evidence about legal meaning rather than about something correlated with it?

3.1 Requirement 1: a measured ceiling

A correlation with human judgment means nothing until one knows what correlation a human achieves. Where a corpus carries $k \geq 3$ ratings per item this is free: hold out one rater, correlate their ratings against the mean of the remaining $k - 1$, and average over the held-out position. Two bounds should be reported, because they answer different questions. The leave-one-annotator-out correlation answers *is the metric as good as an expert?* The Spearman–Brown reliability of the k -rater mean, and its square root, answer *is the metric as good as the aggregate label allows?* A metric can sit above the first and below the second without being superhuman at anything, because predicting an average is easier than being one of the raters averaged: aggregation cancels idiosyncratic variance no individual can cancel alone. Reporting one bound and not the other invites the conflation we document in Section 6.2. Which coefficient is used is itself a substantive choice rather than a formatting one. Rao and Callison-Burch [46] show that for rubric-based judges the scale, the retained cases, the treatment of abstentions and the pooling rule fix what the reported number estimates before any statistic is computed, and that on one benchmark these choices alone move accuracy from 0.551 to 0.899 and carry κ across zero without altering a single verdict. Our nominal-versus-ordinal α discrepancy in Section 6.1 is the same phenomenon in a smaller compass.

3.2 Requirement 2: identical supervision

Comparisons in this literature routinely place an unsupervised cosine similarity and a regressor fine-tuned on the label in the same table, normalise the former by decimal scaling into the label range, and report RMSE for both. Decimal scaling is not calibration: it fixes the range but neither the location nor the shape of the distribution. A cosine similarity occupying $[0.7, 1.0]$ maps to high scores everywhere and records a large RMSE however well it ranks items, so the resulting column measures access to supervision rather than judgment. The repair is to fit a monotone map from each metric’s raw score to the human scale on the training split only, and to evaluate on the held-out split. We use isotonic regression [58], which cannot improve a metric’s ranking because it is monotone. What that monotonicity does and does not buy is worth stating, since it is easy to overclaim. Rank statis-

tics are almost invariant, moving by at most 0.014 across the comparison set, and the residual comes from the ties the fitted step function introduces rather than from any reordering. Pearson r is not invariant, because a monotone map need not be linear: it moves by up to 0.074 here. Calibration is therefore a repair to the location and shape of a metric’s distribution, not a no-op, and the r column of Table 5 is a calibrated quantity like the rest of the table.

3.3 Requirement 3: a trivial-feature control

Simplification changes surface statistics systematically, and the most systematic of them is length: LLM simplifications are usually shorter, and the dominant annotated error category in legal simplification is omission. A rubric deducting score once per identified error therefore produces a label partly predictable from how many words disappeared, with no representation of legal force involved, and any comparison omitting a length control will attribute to a neural metric what belongs to that confound. We require a supervised baseline over surface features alone (word counts, their difference, token and numeral overlap), fitted under Requirement 2 and reported in the same table. The precedent is Pacchiardi et al. [41], whose n -gram classifiers score well on capability benchmarks while possessing none of the capability under test; a benchmark that has not been run against such a baseline has not established that its scores require the construct. The same holds for a metric comparison, where the baseline is cheaper still.

3.4 Requirement 4: dissociation

The three requirements above discipline the corpus-correlation evidence. They cannot establish what a metric is reading, because on naturally occurring pairs surface form and legal force vary together. This is the same confound that makes the identical and unrelated checks uninformative, and it needs the same remedy: an evaluation set on which the two are pulled apart by construction.

Formally, let s' be a perturbation of s with token Jaccard $J(s, s') \approx 1$ and legal force $L(s') \neq L(s)$, and let u be an unrelated sentence of the same register. We report the *margin fraction*

$$\text{margin} = \frac{\overline{m(s, s)} - \overline{m(s, s')}}{\overline{m(s, s)} - \overline{m(s, u)}},$$

the share of the metric’s own identical-to-unrelated dynamic range that it spends on a legally decisive

edit. The normalisation makes the statistic comparable across metrics with incomparable raw scales, and charges each metric against the range it actually has rather than a nominal $[0, 1]$. A metric that treats the flipped sentence as it treats an unrelated one scores 1; a pure overlap function scores near 0.

The margin is not the rate at which a metric ranks (s, s) above (s, s') , and the rate is much the weaker quantity: a metric can rank correctly on every item while moving by a thousandth of its range, a ranking that carries no decision. Nor is the margin redundant with corpus correlation; whether the two agree is an empirical question, and Section 6.5 reports that here the two orderings part at both extremes.

4 LEGAEDIT: Constructing the Dissociation

4.1 Source text and edit inventory

The perturbations need source sentences that are statutory in register, plentiful, and regular enough in phrasing that a rule can apply cleanly. We draw them from two Quebec statutes held out from any corpus used elsewhere in the paper, the Automobile Insurance Act and the Highway Safety Code, and apply a single minimal edit that changes legal force while leaving the sentence otherwise intact. The inventory is organised by legal effect rather than by linguistic category, which is the respect in which it departs from general-domain perturbation suites.

- **Modality:** *doit* $\leftrightarrow$ *peut*; obligation becomes permission.
- **Party:** *l’assuré* $\leftrightarrow$ *l’assureur*; the duty changes hands.
- **Polarity:** dropping *ne ... pas*; *couvre* $\leftrightarrow$ *ne couvre pas*; *exclut* $\leftrightarrow$ *inclut*.
- **Quantum:** the first monetary or temporal amount multiplied by ten (*12 mois* $\rightarrow$ *120 mois*).
- **Scope:** *tous les* $\rightarrow$ *certains*; *sauf si* $\rightarrow$ *si*.
- **Temporal:** *avant* $\leftrightarrow$ *après*.
- **Class** (tier 2): *automobile* $\leftrightarrow$ *véhicule*, the defining example of the construct; and *et* $\leftrightarrow$ *ou*, which converts conjunctive conditions to disjunctive ones.

We separate a *tier 1* set, in which the edit unambiguously changes who is bound, what is covered,

<i>peut</i> → <i>doit</i>	<i>tier 1</i>
<i>o</i> : La Société peut refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
<i>s'</i> : La Société doit refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
<i>doit</i> → <i>peut</i>	<i>tier 1</i>
<i>o</i> : Le candidat doit toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
<i>s'</i> : Le candidat peut toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
<i>drop ne ... pas</i>	<i>tier 1</i>
<i>o</i> : Les feux visés au paragraphe 2° ne sont pas requis pour tout véhicule dont la largeur excède 2,03 mètres.	
<i>s'</i> : Les feux visés au paragraphe 2° sont requis pour tout véhicule dont la largeur excède 2,03 mètres.	
<i>avant</i> ↔ <i>après</i>	<i>tier 1</i>
<i>o</i> : Elle n'est pas due avant le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
<i>s'</i> : Elle n'est pas due après le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
<i>tous les</i> → <i>certains</i>	<i>tier 1</i>
<i>o</i> : Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire toute chose utilisée en contravention au présent article.	
<i>s'</i> : Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire certaine chose utilisée en contravention au présent article.	

Table 1: LEGALEDIT samples, one per perturbation rule, with the edited span in bold. Each remains fluent and statutory in register, shares almost all of its tokens with the original, and states a different rule of law.

or how much, from a *tier 2* set of connective and class shifts, and report both.

4.2 Keeping the edit invisible to a fluency detector

A diagnostic of this kind fails if a metric can find the edited item by noticing that it reads badly. Perturbations introducing a grammatical artefact are therefore filtered by a conservative French well-formedness check covering failed elision (*le automobile*), missing contraction (*de le*), doubled negation and determiner–noun gender disagreement. It flags none of the 400 unmodified source sentences, so it removes only genuine artefacts rather than trimming the set toward any particular rule.

The result is 373 pairs (310 tier 1) with mean token Jaccard 0.93 against their source, none below 0.75. Table 1 gives one example per rule.

4.3 Administering the diagnostic

For each source sentence s with perturbation s' we score three pairs: (s, s) , (s, s') , and (s, u) for an unrelated sentence u drawn from the same statutes. The metric's scores are min–max normalised per metric so that the identical and unrelated probes anchor its range, and the margin fraction of Section 3 is computed from the three means. Because the harness takes an arbitrary scoring function m , the same procedure applies unchanged to untrained metrics, to fine-tuned regressors, and to a prompted LLM judge.

4.4 The comparison set

We administer the diagnostic to French-capable metrics throughout, since scoring English-only metrics on French text confounds language coverage with metric quality. The set covers three families. BERTScore [61] runs over CamemBERTv2 [4] and FlauBERT [36], in both F1 and recall-only variants, since recall directly targets omission. Sentence-embedding cosine uses LaBSE [21], Sentence-CamemBERT, multilingual mpnet and multilingual E5. Bidirectional NLI entailment uses mDeBERTa-XNLI [35]. To these we add token Jaccard and the trivial surface features of Requirement 3, so that the overlap end of the spectrum is represented explicitly rather than by inference.

5 Diagnostic Results: Two Families, Not One

Table 2 reports the margin over 13 scorers, plotted in Figure 3 (Appendix A). The metrics divide cleanly into two families.

Every *similarity*-based metric fails, and fails almost completely. The seven embedding and BERTScore variants spend between 0.022 and 0.039 of their working range on an edit that reverses the law. BERTScore over CamemBERTv2 places a sentence saying the opposite of the original at 0.982, on a scale where the original itself sits at 1.000 and an unrelated statute sits at 0.193. Token Jaccard, at 0.079, is in the same regime, which is the point: these metrics are behaving as overlap functions, exactly as the argument of Section 3 says they must. Their *discrimination* rates are uninformative for the same reason: the seven rank the identical pair above the edited one 100% of the time, and token Jaccard 99.2% of the time, but by a margin so small that it carries no decision. This is the concrete case the margin statistic was defined

to catch.

Bidirectional NLI does not fail. It spends 0.67 of its range on the edit, ranks correctly on 93.3% of items, and places a legally flipped sentence (0.265) nearer the unrelated pair (0.010) than the identical one (0.784). The two directions individually reach 0.424 and 0.473; taking their minimum, which is what an omission-*or*-hallucination reading of the construct requires, is what produces the 0.67. On the property the construct is about, this family is not merely competitive but in a different class.

The prompted judge sits between the two families and belongs to neither. It places the identical pair at 10.00 and the unrelated pair at 1.06 on the rubric’s own ten-point scale, a perfect score on both checks that Section 3 argues are jointly satisfiable by overlap. It nonetheless awards a clause whose legal force has been reversed a mean of 6.69 (6.31 on tier-1 edits), for a margin of 0.370 and correct ranking on 86.3% of items. It is an order of magnitude above the similarity family and well below bidirectional NLI. We take this up in Section 6.4, where it turns out that most of the gap is attributable to the rubric rather than to the model.

Relation to MENLI. That entailment survives meaning-changing edits better than similarity is not a new claim: Chen and Eger [13] establish it for general-domain NLG metrics, and our result confirms it. Three things here are new. The inventory is defined by legal effect rather than by linguistic category, so the diagnostic speaks to a deployment question (may this metric gate a simplified clause?) rather than to metric robustness in general. The dissociation of Section 6.5 is a property of this corpus and could not be read off MENLI. And the check this literature has standardised on would have excluded the entailment family outright, as the next paragraph shows.

The existing checks would have excluded it. Note where NLI’s identical score sits: 0.784, not 1.000. This is not a normalisation artefact. In raw terms mDeBERTa assigns a mean entailment of 0.780 to a sentence paired with *itself*, and clears the 99% bar the identical-pair convention requires on only 1.3% of items, against 100% for every BERTScore and embedding metric. The metric that best tracks legal meaning is therefore the one metric that convention would *disqualify*. We take this as the strongest available argument for the position of Section 3: the identical-pair check as presently applied does not merely fail to identify a legal met-

ric, it actively selects against the family that has the property being sought. A metric that saturates on identical input is easy to build, since every overlap function does it by construction, and saturation is evidence about calibration at one endpoint, not about whether the metric reads the law. It should be reported, and it should not be a filter.

Is the failure uniform across edit types? No rule is reliably detected. The best-detected operation, dropping *ne . . . pas*, still averages only 0.333, and it is plausibly the easiest because it is the one rule that *removes* tokens rather than substituting them. At the other end, swapping the bound party (*l’assuré* for *l’assureur*), which reverses who owes the duty and who is owed it, draws 0.035 over the 14 party-swap items against a mean of 0.155 across all rules, both averaged over the same 13 scorers. Reversing which party a clause binds is close to the least detectable thing one can do to it. Appendix A gives the full breakdown.

6 Case Study: The Protocol on FRJUDGE

The diagnostic answers one of the four requirements. This section runs the other three on FRJUDGE [7], 297 pairs drawn from two Quebec insurance forms, each simplified by gpt-4-turbo-2024-04-09 with a zero-shot prompt and rated by five law students on legal meaning (1–10), simplicity, and an 18-class legal characterization. It then asks whether the two kinds of evidence agree.

Reconstruction. The public `FrJUDGE.jsonl` carries 1,485 rows (297×5) but not annotator identity, which makes Requirement 1 impossible for a downstream user; the raw Prodigy export retains it. We rebuild the corpus from that export, recovering the five annotators (pseudonymised A–E throughout; see the Ethical Considerations section) and the source document of each clause. Our mean pairwise percent agreement is 25.96%, matching the published value exactly, and our per-annotator means (4.33, 7.57, 7.55, 7.72, 4.36) match to within rounding; everything downstream rests on this check.

6.1 Reliability and the structure of disagreement

Krippendorff’s α for legal meaning is 0.104 under the *nominal* coefficient, conventionally the one reported for this corpus, and 0.325 under the *ordinal* one (Table 3). Legal meaning is measured on

Metric	identical	LEGALEDIT	unrelated	discrim.%	margin
NLI-bidirectional (min)	0.784	0.265	0.010	93.3	0.670
NLI-bwd (no omission)	0.784	0.432	0.042	77.2	0.473
NLI-fwd (no hallucination)	0.784	0.469	0.043	77.7	0.424
LLM-judge (deepseek-chat, k=5)	1.000	0.632	0.006	86.3	0.370
surface: token Jaccard	1.000	0.928	0.089	99.2	0.079
LaBSE	1.000	0.972	0.292	100.0	0.039
BERTScore-FlauBERT	1.000	0.979	0.393	100.0	0.035
mE5-base	1.000	0.975	0.266	100.0	0.034
Para-mUSE-mpnet	1.000	0.981	0.312	100.0	0.028
Sentence-CamemBERT	1.000	0.983	0.316	100.0	0.024
BERTScore-CamemBERTv2-Recall	1.000	0.983	0.240	100.0	0.023
BERTScore-CamemBERTv2	1.000	0.982	0.193	100.0	0.022
surface: $- s - o $	1.000	0.997	0.718	11.3	0.011

Table 2: LEGALEDIT. Scores min–max normalised per metric. *discrim.* is the share of items ranked correctly (identical > edited); *margin* is the fraction of the identical-to-unrelated range spent on the legal edit. Correct ranking is necessary but far from sufficient: the similarity metrics rank 100% of items correctly while moving almost nothing, which is why the margin, not the rate, is the quantity to read.

Dimension	Krippendorff’s α		Reported prev.
	nominal	appropriate	
Simplicity level	0.183	0.183	0.18
Characterization	0.552	0.552	0.55
Legal meaning	0.104	0.325	0.10

Table 3: Krippendorff’s α on FRJUDGE. Legal meaning is an ordinal ten-point scale, for which the nominal coefficient is not the appropriate measure. *prev.* is the value reported by Beauchemin et al. [7].

an ordered ten-point scale, and nominal α counts a one-point disagreement as fully as a nine-point one, so the ordinal coefficient is the one appropriate to the design [5, 33]. The nominal values for simplicity (0.183) and characterization (0.552) are genuinely nominal and need no adjustment. The corpus is therefore more reliable than a nominal reading suggests, a point in its favour that a reader would otherwise miss, though 0.325 still indicates only fair agreement.

The disagreement that remains is structural rather than random (Figure 1). The five annotators separate into two stable populations: B, C and D average 7.61 and A and E average 4.34, a gap of over three points on a ten-point scale. Ratings are strongly bimodal (the modal values are 10 and 1), the median across-annotator range on a single pair is 7 points, and 63% of pairs span at least six. Averaging produces a target with a standard deviation of only 2.40, much narrower than the underlying ratings, which is precisely the condition under which a regressor that predicts the middle scores well. The mean of five such ratings is a quantity no annotator asserted, and this is why the

Held-out	r (95% CI)	ρ	RMSE
A	0.524 [0.44, 0.60]	0.551	3.93
B	0.569 [0.48, 0.65]	0.586	3.05
C	0.680 [0.60, 0.75]	0.725	2.75
D	0.601 [0.52, 0.67]	0.557	3.08
E	0.612 [0.55, 0.67]	0.679	3.56
Mean (ceiling)	0.597	0.620	3.27
Best pair (C-D)	0.606	–	–
Mean pairwise	0.462	–	–

Table 4: Leave-one-annotator-out ceiling on FRJUDGE ($n = 297$). A single expert predicts the consensus of the other four at $r = 0.597$.

two bounds below differ so much.

6.2 The human ceiling

Requirement 1 is directly executable here, because every item carries five ratings. We correlate each annotator’s ratings against the mean of the other four (Table 4). The mean leave-one-annotator-out correlation is 0.597 (RMSE 3.27), the best-agreeing pair reaches $r = 0.606$, and the mean pairwise correlation is 0.462.

A model exceeding 0.597 is not thereby super-human: predicting the *mean* of five raters is a different and much easier task than being one of them, and one that rewards predicting the middle of a bimodal distribution. The appropriate upper bound for predicting the observed mean is the reliability of that mean, which by Spearman–Brown from the mean pairwise correlation is 0.811 , so a metric can in principle approach $\sqrt{0.811} = 0.90$. Reporting both bounds separates two claims a single correlation runs together: *as good as an expert* ($r \approx 0.597$) and *as good as the aggregate allows*

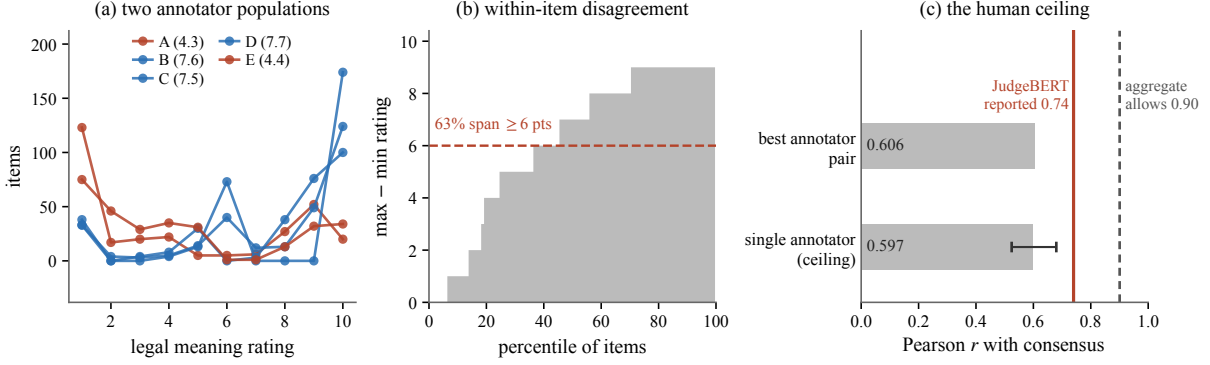


Figure 1: Disagreement in FRJUDGE. (a) The five annotators form two populations: B, C, D (blue) rate legal preservation about three points higher than A and E (red) across the whole corpus. (b) Sorted within-item disagreement; the median item spans 7 points of a ten-point scale. (c) A single annotator predicts the consensus of the other four at $r = 0.597$ (bar; whiskers give the range over the five held-out annotators), and the best-agreeing pair reaches $r = 0.606$.

($r \approx 0.90$). A correlation falling between them is evidence for the second and evidence about nothing at all with respect to the first. Any future legal-meaning metric should report against both.

6.3 Metrics under identical supervision

Requirement 2 asks for a comparison in which every candidate sees the same supervision, and Requirement 3 for a trivial control inside it. We take the comparison set of Section 4.4 and add a supervised ridge regression over ten surface features, mapping every metric’s raw score to the human scale by an isotonic regression fitted on the training split alone.

Three things in Table 5 are worth separating. First, calibration collapses most of the RMSE gap that uncalibrated comparisons record: the metrics were not as far from human judgment as such tables suggest, merely on a different scale. Second, the strongest single unsupervised signal is not a semantic one. Third, a ridge regression over word counts and token overlap reaches $r = 0.62$, at the human ceiling of Section 6.2 and above every metric in the comparison set of Section 4.4. Only two scorers in the table beat it, and neither is a semantic metric: the prompted judge of Section 6.4, and the bare length difference at $r = 0.641$, which is the trivial feature the ridge was built around. Whatever a fine-tuned legal metric adds beyond that, a comparison without this row does not measure it. We read this not as evidence that the corpus is trivial but as evidence about what the *annotation* rewards: score is reduced once per identified error, the most frequent error type in LLM simplification is omission, and omission is strongly correlated with becoming

shorter. A metric can therefore score well by detecting deletion, with no representation of legal force. That hypothesis is what Requirement 4 exists to test.

Statistical adequacy. The test split holds 89 items, at which size differences finer than roughly a quarter of a correlation point are not resolvable, ours included. We report bootstrap confidence intervals throughout [17] and, under the Williams test for dependent correlations, the length feature’s advantage is significant against each of the ten unsupervised metrics in the comparison set (Appendix C); the prompted judge of Section 6.4 is the one scorer that beats it. This establishes that the trivial baseline is genuinely stronger *on this corpus*, not that the neural metrics reason less well about law; distinguishing those two readings is what the diagnostic is for.

6.4 A prompted judge

A prompted model given the annotators’ own rubric [20] is the natural competitor to a fine-tuned 112M-parameter encoder, and its absence was the largest gap in the comparison above. It also brings its own validity problem. Norman et al. [40] evaluate 21 judges over roughly 541,000 judgments and report a consistency-bias paradox, in which test-retest reliability above 0.95 coexists with severe position bias, so a judge can be entirely stable on the checks applied to it and still be measuring the wrong thing. That is the failure mode Section 3 predicts and the one the diagnostic is built to expose. We close the gap with deepseek-chat prompted with the FRJUDGE scale as Section 6

Metric	r	ρ	RMSE _{raw}	RMSE _{cal}	over%
<i>Unsupervised, French-capable</i>					
BERTScore-CamemBERTv2	0.291 $\pm$ 0.09	0.326	2.62	2.36	49.3
BERTScore-CamemBERTv2-Recall	0.352 $\pm$ 0.12	0.398	2.59	2.30	47.2
BERTScore-FlauBERT	0.482 $\pm$ 0.08	0.459	2.16	2.12	49.4
LaBSE	0.409 $\pm$ 0.06	0.402	2.33	2.21	47.2
Sentence-CamemBERT	0.296 $\pm$ 0.05	0.287	2.68	2.31	49.3
Para-mUSE-mpnet	0.252 $\pm$ 0.08	0.266	2.88	2.35	47.2
mE5-base	0.358 $\pm$ 0.08	0.356	2.52	2.26	48.0
NLI-fwd (no hallucination)	0.368 $\pm$ 0.06	0.352	3.85	2.25	49.3
NLI-bwd (no omission)	0.340 $\pm$ 0.09	0.355	3.70	2.28	46.9
NLI-bidirectional (min)	0.395 $\pm$ 0.07	0.405	3.74	2.22	46.6
<i>Trivial surface features</i>					
$- s - o $ (length difference)	0.641 $\pm$ 0.04	0.624	3.25	1.85	47.0
Token Jaccard	0.303 $\pm$ 0.06	0.331	3.34	2.34	48.0
Length ratio	0.580 $\pm$ 0.05	0.560	3.43	1.97	49.1
Surface-Ridge (10 features, supervised)	0.621 $\pm$ 0.05	0.588	1.89	1.89	45.8
<i>Prompted LLM judge</i>					
LLM-judge (deepseek-chat, k=5)	0.755 $\pm$ 0.05	0.753	2.93	1.57	43.9
Human ceiling (one annotator)	0.597	0.620	–	3.27	–

Table 5: Calibrated comparison on the FRJUDGE test split, mean over 10 seeds. RMSE_{raw} is the decimal-scaling protocol in current use; RMSE_{cal} fits an isotonic map on the training split. r and ρ are both computed after calibration; being monotone, the map leaves ρ almost unchanged (at most 0.014) but shifts r by up to 0.074.

describes it: a ten-point ordinal judgment, starting at 10 and deducting one point per identified legal error across the four annotated error types, with readability explicitly excluded. Three properties of the administration matter. Each pair is scored in its own request, so the judge never sees the probe condition, the generating rule, or a neighbouring item. This is a stronger guarantee than shuffling a batch, since a single-item call carries no condition information at all. Each pair is scored $K = 5$ times and the mean reported, because a sampled judge has variance the way training has seed variance. And every response is cached against a hash of model, prompt version, sample index and pair, so the run is reproducible and resumable; the full protocol is 7,080 calls costing \$0.99.

On the corpus, it is the strongest scorer we test. Calibrated under Requirement 2 and evaluated on the held-out splits, the judge reaches $r = 0.755 \pm 0.054$ with RMSE 1.57 (Table 5), above the length feature, above the supervised surface regression, and above our leave-one-annotator-out human ceiling of 0.597. The ceiling caveat of Section 6.2 applies with full force here: it is computed on all 297 items while metric correlations run on 89-item test splits, and it carries its own sampling uncertainty, so this comparison should be read with those intervals in view. The figure is also the least stable quantity we report, since an independent re-

draw moved it from 0.755 to 0.726, and is better quoted as a range than to three digits.

As a sixth annotator, it agrees with the panel while sitting outside it. Because the judge emits on the annotators’ scale it can be placed inside the panel rather than only beside the metrics. Its mean rating is 9.11, against a human panel spanning 4.34 to 7.61, so it is more lenient than any human and far less dispersed. Yet its mean correlation with the five annotators is 0.544, higher than the mean human–human correlation of 0.462. Adding it as a sixth coder nonetheless *lowers* ordinal α from 0.325 to 0.300: it tracks the consensus without belonging to the distribution. Against its own five samples it is far more self-consistent than the humans are with each other, $\alpha = 0.689$ versus 0.325, which complicates the common reading of this construct’s low reliability as irreducible difficulty. Averaging the samples is not cosmetic: correlation against the human mean climbs from 0.611 at $K = 1$ to 0.719 at $K = 5$.

It passes both conventional checks perfectly and still fails the dissociation. On LEGAEDIT the judge scores the identical pair at 10.00 and the unrelated pair at 1.06, which is a flawless result on precisely the two checks Section 3 argues are jointly satisfiable by lexical overlap. It then awards a clause whose legal force has been reversed 6.69 out of ten. Its rubric-level explanations show the

same pattern: on identical pairs it reports an error in 0.0% of samples, and on LEGAEDIT it reports some error in 78.1% and names *incoh rence*, the correct type for a meaning-reversing edit, in 54.6%. On roughly a fifth of law-reversing edits it returns a clean bill of health outright. The nearest published parallel is Xu et al. [53], who evaluate LLM judges against two legal experts on propositions drawn from Court of Justice of the European Union decisions and find that rubric-guided judgment aligns with expert assessment better than direct scoring does, yet remains insensitive to the finer distinctions the experts draw. Their insensitivity is reported qualitatively and on a different jurisdiction, task and rubric; the margin is what puts a number on it.

The rubric, not the model, is what limits it.

That last result is largely an artefact of the annotation scheme, and the demonstration is direct. Holding the model fixed and varying only the wording, the margin moves from 0.421 under the rubric-faithful prompt to 0.687 under a bare instruction that strips the four-type taxonomy and asks only how well the legal meaning is preserved. That is a range of 0.267 across 3 wordings of one question put to one model (Table 6). For scale, an independent redraw of the same prompt moved the margin by 0.002. The bare prompt is better on *both* axes, reaching $r = 0.758$ as well, and it scores the reversed clause at 3.83 against the rubric prompt’s 6.25. The mechanism is visible in the rubric itself: score is reduced *once per identified error*, so a single clause reversal is one incoherence and the scheme caps the penalty near the top of the scale. This is the same property we identify in Section 6.3 as making the label partly predictable from length, reappearing on the diagnostic side. A judge asked to apply the corpus’s own rubric faithfully is thereby prevented from expressing that an edit destroys the legal meaning.

The general form of this result is now documented. Roy et al. [48] treat a rubric as a measurement specification and audit rubric sets for reliability, preference fit and adversarial robustness, finding that no existing source achieves all three at once and that repairing the rubric, rather than changing the model, is what moves judge accuracy. Bellibatlu et al. [9] measure verdict stability across nine judges under semantically equivalent rephrasings of the judge’s own prompt, and report that model scale does not predict consistency. Both

predict the pattern we observe, in which 0.267 of margin turns on the wording of one question. The prompt is the more important experimental variable, and it is the one that metric papers report least.

Capability does not close the gap, and neither does combining.

Two further controls bound the result. Substituting a reasoning model (`deepseek-reasoner`) under the identical rubric and items does not help: $r = 0.580$ against 0.619 and margin 0.320 against 0.425, so the shortfall is not a matter of the model thinking harder. Nor does combination rescue it. We tried six rules for merging the judge with bidirectional NLI, every calibration fitted on the training split alone and the diagnostic never entering the fit. None beats both parents on both axes. The best, a weighted blend at $w = 0.74$, reaches $r = 0.762$ and margin 0.574, gains smaller than the redraw-to-redraw variation. We report this as a negative result rather than proposing a metric.

One judge behaviour survives everything.

Across all 5 configurations, covering both models and all 3 wordings, the judge penalises weakening an obligation far more than strengthening one. On the full diagnostic under the primary configuration, substituting *peut* for *doit* scores 4.72 ($n = 82$) while the reverse substitution scores 8.02 ($n = 111$), a gap of 3.30 that reproduces at 3.02 on the independent redraw. On the 100-edit subsample of Table 6, where every configuration can be compared like for like, the gap ranges only over 3.24–4.03. Both edits change the law by the same construction. A scorer that treats a stricter rewrite as more faithful is tracking something closer to legal *severity* than to legal *equivalence*, and no aggregate margin would reveal it. We regard this as the most robust judge-specific finding here, and as an argument for diagnostics built from directional pairs.

6.5 The two orderings come apart

We can now put the two kinds of evidence side by side. Over the 13 scorers that carry both a calibrated r on FRJUDGE and a LEGAEDIT margin, the set Figure 2 plots, the rank correlation between the two is $\rho = 0.24$ ($p = 0.43$), with a 95% bootstrap interval of $[-0.39, 0.75]$.

That interval is the finding, and it should be read as an absence of resolution rather than as a demonstrated null. We are not entitled to say that corpus

Configuration	r	edit	margin	discrim.%	asym.
chat, rubric (FR) [primary]	0.687	6.25	0.421	82.0	3.24
chat, rubric (EN)	0.727	4.84	0.573	88.0	3.63
chat, bare instruction	0.758	3.83	0.687	89.0	4.03
chat, rubric, $K=1$	0.619	6.22	0.425	79.0	3.34
reasoner, rubric, $K=1$	0.580	7.12	0.320	87.0	3.79

Table 6: The prompted judge across models and prompt wordings, on a fixed 100-edit subsample of LEGAEDIT so the reasoner stays affordable. The first row is the FRJUDGE rubric in French, the second the same rubric in English over the same French text, and the third a bare instruction with the error taxonomy removed. The three wordings are drawn at $K=2$ and the two capability rows at $K=1$; within a row every column comes from the same draw, so the wordings are comparable with each other and the two models with each other, but a row is not comparable with the $K=5$ figures quoted elsewhere. *asym.* is the mean score on a *peut*→*doit* edit minus that on a *doit*→*peut* edit; both reverse the law.

correlation fails to predict legal sensitivity; we are entitled to say that 13 metrics cannot establish that it does. Restricting to the 10 untrained neural baselines, which is the narrower comparison a metric paper would ordinarily make, moves the estimate to $\rho = 0.58$ ($p = 0.08$, 95% CI $[-0.08, 0.90]$), a moderate positive value pointing the other way. We report both because the difference between them is instructive: the subset excludes the length feature and the prompted judge, and those are precisely the two scorers at which the orderings part. A coefficient computed after removing the disagreeing cases is not evidence of agreement.

The extremes are the durable part of this result, and they do not depend on a rank correlation over thirteen points. The trivial length feature is the best-correlating signal on the corpus ($r = 0.641$) and has the lowest margin of anything we test (0.011, ranking the edited pair correctly only 11.3% of the time). Bidirectional NLI sits mid-table on correlation ($r = 0.395$) and is the only family that reacts to the edit at all. A metric selected on corpus correlation alone would pick the wrong one. The prompted judge makes the same point from the other end: it is the strongest scorer in the comparison on the corpus ($r = 0.755$) and fourth of 13 on the diagnostic, so the two orderings disagree at both extremes rather than only at the bottom. Figure 2 plots the two axes against each other, and the emptiness of its upper-right quadrant is the finding.

6.6 Implications for training a metric

The protocol also constrains how one would fit a new metric. Appendix B reports seven CamemBERTv2 variants trained on identical splits and evaluated on the human-annotated test split only, never on a split containing their own augmentation. Three results bear on the protocol.

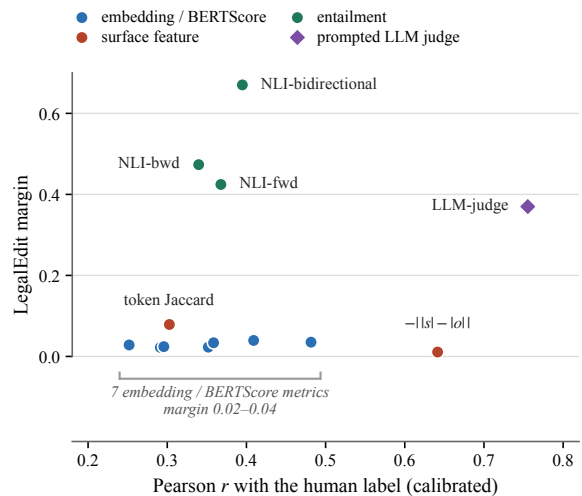


Figure 2: The two orderings come apart. Horizontally, how well a metric correlates with the human label on FRJUDGE after calibration, the quantity a conventional comparison selects on. Vertically, the share of its working range it spends on an edit that reverses the law. Selecting on the horizontal axis picks the prompted judge and then the length feature, the two points on the right; the length feature is the worst scorer of all on the vertical axis. The entailment family, the only untrained one that moves appreciably, sits in the middle of the horizontal axis. The upper-right quadrant is empty.

First, no training target moves the correlation by more than the scalar baseline’s own spread across seeds (± 0.082). What the disagreement-aware objectives buy is stability and calibration instead: the quantile head overshoots the human label on 9.9% of items against 25.4% for the scalar head, so risk asymmetry can be written into the loss rather than hoped for, at no cost in correlation.

Second, training on a constraint buys satisfaction of it in distribution and nothing across a change of source. Every model, including the two whose mixture contains identical pairs, scores 0% on an

identical-pair probe from a different document, and DA+LEGALEDIT, trained on minimally-edited negatives, reaches a margin of 0.051 against 0.67 for untrained bidirectional NLI.

Third, scored on human items alone, the identical/unrelated augmentation costs more than half the correlation on five seeds of five. A figure computed on a split containing that augmentation is dominated by identical pairs at 10 and unrelated pairs at 1 supplying most of the label variance, so separating the two evaluations is Requirement 2 applied to training rather than to baselines.

7 Discussion

Three checks, none of them a filter. A metric claiming to measure legal meaning should report three checks, not two: identical pairs (does it saturate correctly?), unrelated pairs (does it bottom out?), and LEGALEDIT (does it move when, and only when, the law moves?). The first two are jointly satisfiable by lexical overlap and so cannot distinguish a legal metric from a similarity function; the third cannot be passed by overlap at all. That LEGALEDIT is passable matters as much as that it is hard: because bidirectional NLI clears it while every similarity metric does not, a low margin identifies a fixable deficiency rather than an inherent limit of automatic evaluation.

The three should be reported *jointly and separately*, and none used as an admission criterion. Here the metric that scores highest on the diagnostic fails the identical-pair check most badly, and a pipeline screening candidates on saturation, as the current convention invites, would have discarded it before the interesting question was asked. Saturation, bottoming out and legal sensitivity are three independent properties; a scorer can be repaired for the first two by calibration and cannot be repaired for the third at all. Rank on the property that cannot be fixed. Nor has a metric trained on its own sanity checks thereby passed them: our augmented models pass the unrelated check on 77% and 69% of probes and the identical check on none, and the only difference is whether the probe comes from the same documents as the training pairs (Section 6.6). The honest report separates “satisfies the constraint by construction” from “generalises to the constraint”.

Entailment is the better starting point. The construct FRJUDGE annotates is defined by an error taxonomy whose two dominant categories are

omission and hallucination, precisely the two directions of entailment failure. The only scorer that reacts to a legally decisive edit is the one built from bidirectional entailment. We would not conclude that an off-the-shelf multilingual NLI model is an adequate legal metric: its correlation with the human label is middling ($r = 0.395$), it is slower than a fine-tuned encoder, and its margin of 0.67 leaves a third of the range unspent. The narrower conclusion is that the next legal-meaning metric should be built on an entailment objective and evaluated on the diagnostic, rather than on a similarity objective selected by corpus correlation; a supervised model trained against FRJUDGE *and* minimally-edited negatives is the obvious candidate. It is a starting point and not a destination. The failure You and Guo [57] document in biomedical plain language summarisation has a direct legal analogue: a simplification that glosses a defined term for a lay reader adds content the clause does not contain, and the $o \Rightarrow s$ direction of our bidirectional score will read that as hallucination. Repairing that direction, rather than adopting entailment unmodified, is what the recommendation amounts to.

Disagreement is the finding, not the noise. That two of five Quebec law students systematically rate legal preservation three points lower than the other three is not an annotation defect to be averaged away. It measures how much legal interpretation of an isolated clause varies among trained readers, and any deployed metric returning a single number conceals it. This is the reading the perspectivist literature now takes as default [55], and the practical consequence Bonagiri et al. [10] draw, that rankings computed on a collapsed label are unstable across annotator subsets, applies to any future leaderboard built on this corpus. A metric that reports “6.2, experts ranged 2–9” tells a practitioner that this clause needs a human, which “6.2” does not.

8 Conclusion

Legal meaning preservation is the right construct to evaluate, and the resources that now exist for it deserve to be built on. But the evidence a metric paper in this area conventionally offers cannot establish that the metric measures legal meaning rather than lexical overlap and length: a correlation with an averaged human label, computed against uncalibrated baselines, without a trivial control, on sanity checks the model may have been trained on.

The missing ingredient is a dissociation, and it is cheap: hold surface form fixed and move the law. On that test every similarity-based metric we examined scores a sentence saying the opposite of the original as though nothing had changed, and reversing which party a clause binds is among the least detectable edits of all. The test is not unpassable. Bidirectional entailment, a family the standard comparison leaves out and the standard sanity check would have disqualified, clears most of it, which points at what the next version of such a metric should be made of. It also carries a warning about how these metrics are selected, because a bare length feature outscores every metric in the comparison set on the corpus and ranks last of all on the law.

Limitations

Our LEGAEDIT labels are definitional rather than annotated: we assert that changing *doit* to *peut* changes legal meaning, rather than having lawyers rate each perturbation. We think this is defensible for tier-1 edits, which alter modality, party, polarity or quantum, and we report tier-2 scope shifts separately for exactly this reason; but a human-validated version would be stronger, and a small expert study is the obvious next step. The perturbations are also single-edit and template-generated, so they probe a specific and narrow kind of legal change. A system could in principle be tuned to them without acquiring general legal sensitivity, which is why we treat LEGAEDIT as a necessary condition and not a benchmark to be optimised.

That caveat bears directly on our most encouraging result, and we would rather state it than let a reader find it. Our perturbations are largely negation, modality, quantifier and antonym contrasts, which is close to the inventory natural language inference training data is built around. A model fine-tuned on XNLI has seen a great many *ne . . . pas* insertions and *tous/certains* alternations, so bidirectional NLI’s margin of 0.67 may reflect familiarity with the *form* of our edits rather than sensitivity to their legal consequence. The two hypotheses make different predictions: only the second should survive perturbations that change legal force without changing logical form, such as substituting a defined term for a near-synonym that is not its legal equivalent. Separating them needs a version of the diagnostic built around defined-term substitution, which our rule inventory reaches only through the

single *automobile/véhicule* case. We report the NLI result as a promising direction with an identified confound, not as evidence that entailment models read law. The prompted judge narrows this without settling it: it has no XNLI fine-tuning and still clears the similarity family by an order of magnitude, so the diagnostic is measuring something available to an architecture without that training history. It does not reach bidirectional NLI’s margin under the same rubric either, so it does not explain NLI’s lead away.

A related objection cuts at the design rather than at the model, and we cannot fully answer it either. Yang et al. [56] argue that any counterfactual edit is a compound treatment, bundling the variable of interest with incidental surface variation, so an observed effect belongs to the target factor only once it has been compared against a meaning-preserving paraphrase baseline; on MEDQA a 14.9% flip rate under a demographic intervention proved statistically indistinguishable from the 14.1% that paraphrasing alone produces. Our margin carries a partial defence, since it is normalised against each metric’s own identical-to-unrelated range and the identical probe is a null-edit anchor, while the well-formedness filter bounds incidental variation and a single substitution leaves 0.93 of tokens in place. But we run no paraphrase-control arm. For the similarity metrics little turns on it, since margins this close to zero leave nothing for a control to explain away; the objection bites on bidirectional NLI, where it compounds with the XNLI confound above. Both point to the same fix, so we state it once: the next version of the diagnostic should carry legally inert paraphrases and defined-term substitutions alongside the legally decisive edits.

A further mismatch is that the diagnostic and the correlation evaluation run on different text. FR-JUDGE is drawn from insurance forms, while the LEGAEDIT probes are generated from two Quebec statutes, because the statutes supply the volume and the regularity of phrasing that the perturbation rules need. The two are the same language and register but not the same genre, so Table 2 and Table 5 answer different questions about the same metric and should not be read as one ranking.

Our prompted-judge results rest on a single model from a single vendor. We ran two controls within reach, a reasoning model under the identical rubric and 3 prompt wordings. The first moved the result very little while the second moved it a great deal, which locates the risk precisely: a second *ven-*

dor would most likely shift the numbers less than a second *prompt* already does. It would still be worth having, and we could not afford it. The prompt finding also means Section 6.4’s placement of the judge below bidirectional NLI is conditional on administering the corpus’s own rubric. We think that is the right primary choice: the rubric defines the construct FRJUDGE operationalises, and a judge asked to predict those labels should be given those instructions. Under a bare instruction, however, the same model reaches 0.687, above bidirectional NLI’s 0.67, and a reader who prefers that framing should have the number. We report the rubric-faithful configuration throughout and the full spread in Table 6.

Two further judge caveats. FRJUDGE is publicly released and both source statutes are public, so contamination cannot be excluded; the perturbations themselves are newly generated, which is the defence, but not a guarantee. And our rubric prompt is reconstructed from the published description of the annotation rather than from the annotators’ actual instructions, which are not in the release. Given how far 0.267 of margin turns on wording, that reconstruction is a load-bearing assumption rather than a detail.

Like the corpus we build on, we evaluate isolated clauses out of document context, and all simplifications come from a single generator (gpt-4-turbo) with one prompt. A metric meta-evaluation properly requires outputs from several systems spanning a range of quality; with one generator, a metric may be learning that generator’s characteristic failure modes. Appendix D addresses document-level leakage but cannot address generator-level overfitting.

Finally, our human ceiling is computed on all 297 items while the model correlations are computed on 89-item test splits, and the ceiling carries its own sampling uncertainty (per-annotator 95% CIs in Table 4). The comparison “model > ceiling” should be read with those intervals in view.

Ethical Considerations

Annotator privacy. FRJUDGE’s five annotators are pseudonymised as A–E in the paper that introduces them, and we use those pseudonyms throughout. Our released code pseudonymises annotator identifiers on load, and no annotator name appears in any result file, table or figure we release. We also ship the pseudonymisation script separately so that anyone redistributing a reconstruction of the

corpus can apply it before doing so.

Use of the metric. Our results strengthen a caution Beauchemin et al. [7] themselves raise. A metric that cannot distinguish *doit* from *peut* must not be used as an automated gate on whether a simplified clause is safe to publish. We regard the appropriate deployment as triage, that is, the ranking of clauses for human review and the flagging of disagreement, rather than as certification. Nothing here constitutes legal advice.

Source material. The statutes from which LEGALEDIT is generated are public legal texts. The perturbed sentences are deliberate misstatements of Quebec law and are released as an evaluation artefact only; they are labelled as such in the distribution and should not be quoted as statements of the law.

References

- [1]. Sweta Agrawal and Marine Carpuat. 2024. Do text simplification systems preserve meaning? a human evaluation via reading comprehension. *TACL*, 12:432–448.
- [2]. Fernando Alva-Manchego, Louis Martin, Antoine Bordes, Carolina Scarton, Benoît Sagot, and Lucia Specia. 2020. ASSET: A dataset for tuning and evaluation of sentence simplification models with multiple rewriting transformations. In *ACL*.
- [3]. Miriam Anschutz, Diego Miguel Lozano, and Georg Groh. 2023. This is not correct! negation-aware evaluation of language generation systems. In *INLG*, pages 163–175.
- [4]. Wissam Antoun, Francis Kulumba, Rian Touchent, Éric de la Clergerie, Benoît Sagot, and Djamé Seddah. 2024. CamemBERT 2.0: A smarter French language model aged to perfection. *arXiv preprint arXiv:2411.08868*.
- [5]. Ron Artstein and Massimo Poesio. 2008. Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.
- [6]. Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry Mayne, Jan Bartzner, Negar Foroutan, Chris Schmitz, Karolina Korgul, Hunar Batra, Oishi Deb, Emma Beharry, Cornelius Emde, Thomas Foster, Anna Gausen, María Grandury, Simeng Han, Valentin Hofmann, Lujain Ibrahim, and 23 others. 2025. Measuring what matters: Construct validity in large language model benchmarks. In *Advances in Neural Information Processing Systems 39 (NeurIPS) Datasets and Benchmarks Track*.

- [7]. David Beauchemin, Michelle Albert-Rochette, Richard Khoury, and Pierre-Luc Déziel. 2025. JUDGEBERT: Assessing legal meaning preservation between sentences. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 92–118.
- [8]. David Beauchemin, Horacio Saggion, and Richard Khoury. 2023. MeaningBERT: assessing meaning preservation between sentences. *Frontiers in Artificial Intelligence*, 6:1223924.
- [9]. Rohith Reddy Bellibatlu, Edward Raff, and Wenbin Zhang. 2026. JudgeSense: A benchmark for prompt sensitivity in LLM-as-a-Judge systems. *arXiv preprint arXiv:2604.23478*.
- [10]. Akash Bonagiri, Gerard Janno Anderias, Saeed Patil, Angelina Lai, Devang Borkar, Gezheng Kang, Ishant Gandhi, Setareh Rafatirad, and Houman Homayoun. 2026. STABLEVAL: Disagreement-aware and stable evaluation of AI systems. *arXiv preprint arXiv:2605.02122*.
- [11]. Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With little power comes great responsibility. In *EMNLP*, pages 9263–9274.
- [12]. Rémi Cardon and Natalia Grabar. 2020. French biomedical text simplification: When small and precise helps. In *COLING*, pages 710–716.
- [13]. Yanran Chen and Steffen Eger. 2023. MENLI: Robust evaluation metrics from natural language inference. *TACL*, 11:804–825.
- [14]. Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *ACL*.
- [15]. Liam Cripwell, Joël Legrand, and Claire Gardent. 2024. Evaluating document simplification: On the importance of separately assessing simplicity and meaning preservation. In *Proceedings of the 3rd Workshop on Tools and Resources for People with READING Difficulties (READI) @ LREC-COLING 2024*, pages 1–14.
- [16]. Ido Dahan, Omer Toledano, Roey J. Gafter, Sharon Pardo, Oren Tsur, Hila Zahavi, and Elior Sulem. 2026. Translate or simplify first: An analysis of cross-lingual text simplification in English and French. *arXiv preprint arXiv:2604.23844*.
- [17]. Daniel Deutsch, Rotem Dror, and Dan Roth. 2021. A statistical analysis of summarization evaluation metrics using resampling methods. *TACL*, 9:1132–1146.
- [18]. Athiya Deviyani and Fernando Diaz. 2025. Contextual metric meta-evaluation by measuring local metric accuracy. In *Findings of the Association for Computational Linguistics: NAACL*.
- [19]. Sri Harsha Dumpala, Aman Jaiswal, Chandramouli Sastry, Evangelos Milios, Sageev Oore, and Hassan Sajjad. 2024. VISLA benchmark: Evaluating embedding sensitivity to semantic and lexical alterations. *arXiv preprint arXiv:2404.16365*.
- [20]. Joseph Enguehard, Morgane Van Ermenegem, Kate Atkinson, Sujeong Cha, Arijit Ghosh Chowdhury, Prashanth Kallur Ramaswamy, Jeremy Roghair, Hannah R. Marlowe, Carina Suzana Negreanu, Kitty Boxall, and Diana Mincu. 2025. LeMAJ (legal LLM-as-a-judge): Bridging legal reasoning and LLM evaluation. In *Proceedings of the Natural Legal Language Processing Workshop 2025*, pages 318–337.
- [21]. Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. In *ACL*.
- [22]. Aparna Garimella, Abhilasha Sancheti, Vinay Aggarwal, Ananya Ganesh, Niyati Chhaya, and Nandakishore Kambhatla. 2022. Text simplification for legal domain: Insights and challenges. In *Natural Legal Language Processing Workshop*, pages 296–304.
- [23]. Natalia Grabar and Rémi Cardon. 2018. CLEAR—simple corpus for medical French. In *Workshop on Automatic Text Adaptation*, pages 3–9.
- [24]. Yvette Graham and Timothy Baldwin. 2014. Testing for significance of increased correlation with human judgment. In *EMNLP*.
- [25]. Yoav Gur-Arieh, Ana Marasović, and Mor Geva. 2026. Faithfulness metrics don’t measure faithfulness: A meta-evaluation with ground truth. *arXiv preprint arXiv:2605.25052*.
- [26]. Margaret Hagan. 2023. Good AI legal help, bad AI legal help: Establishing quality standards for responses to people’s legal problem stories. *SSRN*. SSRN working paper 4696936.
- [27]. Emaan Hariri and Daniel E. Ho. 2025. AI for statutory simplification: A comprehensive state legal corpus and labor benchmark. In *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAAIL)*.
- [28]. Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. CUAD: An expert-annotated NLP dataset for legal contract review. In *NeurIPS Datasets and Benchmarks*.
- [29]. Yiran Hu, Huanghai Liu, Chong Wang, Kunran Li, Tien-Hsuan Wu, Haitao Li, Xinran Xu, Siqing Huo, Weihang Su, Ning Zheng, Siyuan Zheng, Qingyao Ai, Yun Liu, Renjun Bian, Yiqun Liu, Charles L. A. Clarke, Weixing Shen, and Ben Kao. 2026. Evaluation of large language models in legal applications: Challenges, methods, and future directions. *arXiv preprint arXiv:2601.15267*.

- [30]. Tannon Kew, Alison Chi, Laura Vázquez-Rodríguez, Sweta Agrawal, Dennis Aumiller, Fernando Alva-Manchego, and Matthew Shardlow. 2023. BLESS: Benchmarking large language models on sentence simplification. In *EMNLP*, pages 13291–13309.
- [31]. Roger Koenker and Gilbert Bassett Jr. 1978. Regression quantiles. *Econometrica*, 46(1):33–50.
- [32]. Maria Korobeynikova, Alessia Battisti, Lukas Fischer, and Yingqiang Gao. 2025. DETECT: Determining ease and textual clarity of German text simplifications. *arXiv preprint arXiv:2510.22212*.
- [33]. Klaus Krippendorff. 2004. *Content Analysis: An Introduction to Its Methodology*. Sage.
- [34]. Philippe Laban, Tobias Schnabel, Paul N Bennett, and Marti A Hearst. 2022. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *TACL*, 10:163–177.
- [35]. Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. 2024. Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*, 32(1):84–100.
- [36]. Hang Le, Loïc Vial, Jibril Frej, Vincent Segonne, Maximin Coavoux, Benjamin Lecouteux, Alexandre Allauzen, Benoît Crabbé, Laurent Besacier, and Didier Schwab. 2020. FlauBERT: Unsupervised language model pre-training for French. In *LREC*.
- [37]. Mounica Maddela, Yao Dou, David Heineman, and Wei Xu. 2023. LENS: A learnable evaluation metric for text simplification. In *ACL*.
- [38]. Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D. Manning, and Daniel E. Ho. 2025. Hallucination-free? Assessing the reliability of leading AI legal research tools. *Journal of Empirical Legal Studies*, 22(2):216–242.
- [39]. Zain Muhammad Mujahid, Dustin Wright, and Isabelle Augenstein. 2026. Stress testing factual consistency metrics for long-document summarization. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*.
- [40]. Justin D. Norman, Michael U. Rivera, and D. Alex Hughes. 2026. Reliability without validity: A systematic, large-scale evaluation of LLM-as-a-Judge models across agreement, consistency, and bias. *arXiv preprint arXiv:2606.19544*.
- [41]. Lorenzo Pacchiardi, Marko Tesic, Lucy G. Cheke, and José Hernández-Orallo. 2024. Leaving the barn door open for Clever Hans: Simple features predict LLM benchmark answers. *arXiv preprint arXiv:2410.11672*.
- [42]. Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In *ACL*, pages 311–318.
- [43]. Joshua C Peterson, Ruairidh M Battleday, Thomas L Griffiths, and Olga Russakovsky. 2019. Human uncertainty makes classification more robust. In *ICCV*.
- [44]. Manuel Pita. 2026. Correct codes for the wrong reasons? Validating LLMs as measurement instruments for theoretical constructs. *arXiv preprint arXiv:2606.28574*.
- [45]. Jipeng Qiang, Minjiang Huang, Yi Zhu, Yunhao Yuan, Chaowei Zhang, and Kui Yu. 2025. Redefining simplicity: Benchmarking large language models from lexical to document simplification. *arXiv preprint arXiv:2502.08281*.
- [46]. Delip Rao and Chris Callison-Burch. 2026. Agreement metrics for LLM-as-Judge evaluation: What to report and why. *arXiv preprint arXiv:2606.00093*.
- [47]. Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *EMNLP-IJCNLP*.
- [48]. Swastik Roy, Rajkumar Pujari, Tharindu Kumarage, Charith Peris, Rahul Gupta, Anna Rumshisky, Pradeep Natarajan, and Venkatesh Saligrama. 2026. PReMISE: Policy rubrics as measurement specifications for LLM judges. *arXiv preprint arXiv:2605.30803*.
- [49]. Ananya B Sai, Tanay Dixit, Dev Yashpal Sheth, Sreyas Mohan, and Mitesh M Khapra. 2021. Perturbation CheckLists for evaluating NLG evaluation metrics. In *EMNLP*, pages 7219–7234.
- [50]. Elior Sulem, Omri Abend, and Ari Rappoport. 2018. BLEU is not suitable for the evaluation of text simplification. In *EMNLP*, pages 738–744.
- [51]. Jan Trienes, Sebastian Joseph, Jörg Schlötterer, Christin Seifert, Kyle Lo, Wei Xu, Byron C Wallace, and Junyi Jessy Li. 2024. InfoLossQA: Characterizing and recovering information loss in text simplification. In *ACL*, pages 4263–4294.
- [52]. Alexandra N Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *JAIR*, 72:1385–1470.
- [53]. Shanshan Xu, Johan Lindholm, Amogh Raina, Henrik Palmer Olsen, and Daniel Hershcovich. 2026. LP-Eval: Rubric and dataset for measuring the quality of legal proposition generation. *arXiv preprint arXiv:2605.19815*.
- [54]. Wei Xu, Courtney Napoles, Ellie Pavlick, Quanze Chen, and Chris Callison-Burch. 2016. Optimizing statistical machine translation for text simplification. *TACL*, 4:401–415.

- [55]. Yinuo Xu and David Jurgens. 2026. Beyond consensus: Perspectivist modeling and evaluation of annotator disagreement in NLP. *arXiv preprint arXiv:2601.09065*.
- [56]. Zihao Yang, Mosh Levy, Yoav Goldberg, and Byron C. Wallace. 2026. Compared to what? Baselines and metrics for counterfactual prompting. In *Conference on Language Modeling (COLM)*.
- [57]. Zhiwen You and Yue Guo. 2026. PlainQAFact: Retrieval-augmented factual consistency evaluation metric for biomedical plain language summarization. *Journal of Biomedical Informatics*, 178:105019.
- [58]. Bianca Zadrozny and Charles Elkan. 2002. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699.
- [59]. Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *ACL*, pages 11328–11348.
- [60]. Leixin Zhang and Çağrı Çöltekin. 2026. Quantifying and predicting disagreement in graded human ratings. In *Proceedings of the 5th Workshop on Perspectivist Approaches to NLP (NLPerspectives) at LREC*.
- [61]. Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating text generation with BERT. In *ICLR*.

A Supplementary Tables and Figures

Margin by perturbation rule. Table 7 gives the per-rule breakdown that Section 5 summarises, averaged over metrics. Two details behind the party-swap figure quoted there are worth stating. The two directions of the swap draw mean margins of 0.046 and -0.005 on 11 and 3 items respectively, which is why we pool them rather than report either cell on its own: over all 14 party-swap items the seven similarity metrics spend 0.020 of their range, with a per-metric spread of 0.006–0.036, against 0.029 for those same seven metrics across all rules. The contrast is milder on this restricted basis than the all-metric figures in Section 5 suggest, because the similarity family has almost no range to spend on any rule. We also tested whether the residual ordering across rules is explained by the size of the edit in characters, and found no relationship ($r = 0.25$ over the 12 rules, 95% CI $[-0.66, 0.73]$). With 12 rules of unequal frequency the ordering is not itself well determined, so we report it descriptively; the

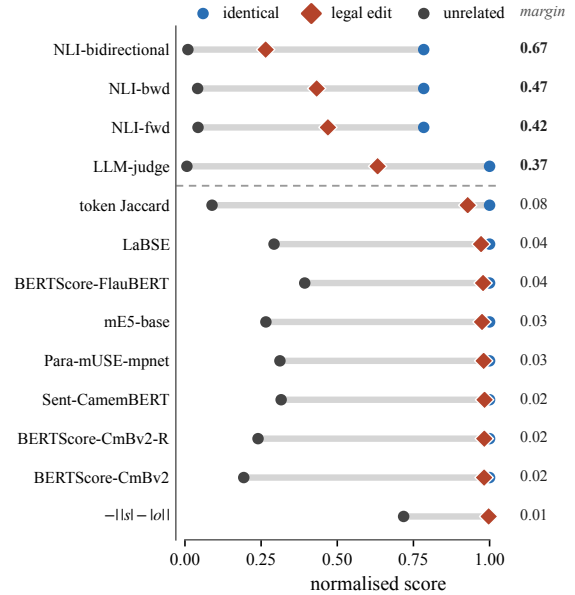


Figure 3: LEGAEDIT. For each metric the grey bar spans its score on an unrelated pair (dark circle) to its score on an identical pair (blue circle); its full working range. The red diamond is where it places a sentence whose legal force has been reversed by a single word. A metric that measured legal meaning would put the diamond at the left end. Every similarity-based metric puts it at the right, hard against the identical pair. Only the four scorers above the dashed rule, the three NLI variants and the prompted judge, move it appreciably. Metric names are abbreviated; Table 2 gives them in full, with the margin column repeated at the right here.

robust observation is the ceiling on detectability, not the ranking.

B Training Variants in Full

The protocol so far audits scorers that already exist. It also has implications for how one would fit a new one, and this section tests two of them. The first follows from Section 6.1: the gold label is the mean of five ratings whose median spread is 7 points, so fitting a scalar regressor to that mean asks the model to predict a quantity none of the annotators asserted, and rewards predicting the centre. The second follows from Section 3: if a diagnostic identifies a deficiency, the obvious response is to train against it.

The seven configurations share an encoder (CamemBERTv2-base), splits, schedule and seeds, and differ only in the target and the loss.

- **Scalar** (the conventional objective): MSE against the mean rating.
- **Distributional**: a ten-way head trained with

Perturbation	<i>n</i>	mean margin	worst
drop <i>ne ... pas</i>	21	0.333	0.032
<i>avant</i> ↔ <i>après</i>	9	0.272	0.006
amount × 10	21	0.256	0.025
<i>doit</i> → <i>peut</i>	82	0.241	0.015
<i>sauf si</i> → <i>si</i>	10	0.225	0.014
<i>peut</i> → <i>doit</i>	111	0.167	0.018
<i>tous les</i> → <i>certains</i>	42	0.153	0.017
<i>et</i> → <i>ou</i>	27	0.069	0.007
<i>ou</i> → <i>et</i>	26	0.069	0.004
<i>assureur</i> → <i>assuré</i>	11	0.046	0.007
<i>automobile</i> ↔ <i>véhicule</i>	10	0.027	-0.025
<i>assuré</i> → <i>assureur</i>	3	-0.005	-0.086
All	373	0.155	–

Table 7: LEGAEDIT margin by perturbation rule, averaged over the metrics in Table 2. *worst metric* is the lowest margin any single metric assigns that rule. A margin of 1 would mean the metric treats a legally flipped sentence exactly as it treats an unrelated one.

KL divergence against the empirical rating histogram, following Uma et al. [52] and Peterson et al. [43]. The reported score is the distribution’s expectation, so it is directly comparable, but the model additionally exposes its spread.

- **Annotator-aware:** five regression heads, one per annotator, averaged at inference. This lets the model represent the lenient and strict populations of Section 6.1 separately instead of averaging them away.
- **Quantile:** a pinball loss [31] at $\tau = 0.25$. The asymmetry is deliberate. A metric that *over-shoots* the human rating is permissive of bad simplifications, while one that *undershoots* is merely strict, and in a legal setting those errors are not symmetric; a quantile objective encodes that preference in the loss rather than hoping it emerges.
- **Multi-task:** the scalar head plus an auxiliary characterization head, trained against the majority vote of the five annotators, which spans 17 of the rubric’s 18 classes (the eighteenth never carries a majority). The characterization labels are collected in FRJUDGE and then discarded; since the published rubric has annotators characterize a clause *before* scoring it, the label plausibly carries information about the score.
- **DA:** the scalar objective with the identical/unrelated augmentation in current use added to the training mixture, unchanged.

- **DA+LegalEdit:** DA plus minimally-edited negatives generated from FRJUDGE originals by the rules of Section 4, disjoint from the diagnostic’s own sentences.

Evaluation protocol. Correlation and RMSE are computed *only* on the human-annotated test split, never on a split containing augmentation. This matters: adding identical pairs (label 10) and unrelated pairs (label 1) to the evaluation set inflates label variance at the extremes, which raises Pearson r for every metric without any improvement in judgment. A table that does not separate the two shows the signature clearly, in that a metric’s r and its RMSE improve and worsen together rather than moving as one. Augmentation here affects training only; the three diagnostic regimes are reported separately as pass rates.

Disagreement-aware objectives buy stability, not accuracy. Table 8 reports the grid, and the first result is a negative one: no training target moves the correlation by more than the scalar baseline’s own spread across seeds (± 0.082), which is itself smaller than the 0.73 gap Section 6.3 showed to be resolvable on an 89-item split. The differences that survive are in the other columns. The distributional head is markedly steadier than the scalar head it replaces (± 0.029 against ± 0.082 over the same five seeds), reaches the best RMSE in the table (1.99), and roughly doubles the rank correlation ($\rho = 0.578$ against 0.280), which is what one expects from a target carrying the shape of the rating distribution and not only its mean. The quantile head does precisely what it was asked to do: it overshoots the human label on 9.9% of test items against 25.4% for the scalar head, the lowest rate of any model here. The risk asymmetry a legal setting calls for can therefore be written into the loss rather than hoped for, and it costs nothing in correlation.

Trained constraints do not cross a change of document. Every model in Table 8 scores 0% on the identical-pair check, the two augmented ones included, although identical pairs are in their training mixture; the augmented models pass the unrelated check on 77% and 69% of probes and the un-augmented models on none. The asymmetry is the one our probe design predicts. Our unrelated probes are drawn from the same two statutes the training mixture draws its unrelated pairs from, so passing that check is an in-distribution result; our

Model	r	ρ	RMSE	over%	ident.	unrel.	LEGAEDIT	
					pass	pass	false-acc.	margin
JudgeBERT-Scalar (baseline)	0.542 $\pm$ 0.08	0.280	2.48	25.4	0	0	0	–
JudgeBERT-Dist (soft labels)	0.572 $\pm$ 0.03	0.578	1.99	42.9	0	0	69	0.015
JudgeBERT-Annot (5 heads)	0.534 $\pm$ 0.05	0.585	3.47	12.6	0	0	0	–
JudgeBERT-Quantile (tau=.25)	0.553 $\pm$ 0.06	0.377	3.28	9.9	0	0	0	–
JudgeBERT-MT (+character.)	0.519 $\pm$ 0.06	0.232	2.51	26.3	0	0	0	–
JudgeBERT-DA	0.192 $\pm$ 0.25	0.112	2.31	42.5	0	77	18	0.019
JudgeBERT-DA+LegalEdit	0.122 $\pm$ 0.10	0.174	2.60	35.3	0	69	0	0.051
<i>NLI-bidirectional</i> (untrained)	0.395	0.405	2.22	46.6	1	–	–	0.670
Human ceiling	0.597	0.620	3.27	–	–	–	–	–

Table 8: Trained metrics, mean over 5 seeds, evaluated on the human-annotated test split only. *ident.* and *unrel.* are the two sanity checks in current use; LEGAEDIT false-acceptance is the share of legally flipped pairs still scored ≥ 7 (“accurate” under the published rubric). The margin is the quantity defined in Section 3. It is shown as ‘–’ where a model places the identical and unrelated probes within one point of each other, because a model with no working range has none to spend on the edit.

identical probes are statutory sentences while the mixture’s identical pairs are FRJUDGE clauses, so passing that one would require generalising across source documents, and no model does. The same holds for the diagnostic: DA+LEGAEDIT, trained on minimally-edited negatives, reaches a margin of 0.051 against 0.67 for untrained bidirectional NLI, its negatives generated from FRJUDGE originals and its probes from statutes. Training on a constraint buys satisfaction of it in distribution and nothing at all across a change of source.

The same separation explains the augmented mixtures’ correlations, which are the clearest effect in the table: DA loses 0.350 against the un-augmented baseline and DA+LEGAEDIT 0.420, on five seeds out of five in both cases, and DA varies by ± 0.254 across seeds, reaching $r = -0.188$ on one of them.

One column of Table 8 should be read with care. LEGAEDIT false-acceptance is the share of flipped pairs still scored ≥ 7 , so it is confounded with where a model puts its predictions overall. The distributional head’s 69% reflects a head that scores high in general (its identical probes average 7.22), and the 0% of the scalar heads reflects models that score low on everything, not models that reject the edit.

C Williams Tests Against the Length Baseline

The test split holds 89 items, at which size the smallest correlation reliably distinguishable from $r = 0.46$ at 80% power is 0.73. The ten unsupervised metrics and the length feature of Table 5

correlate with the *same* human label on the *same* 297 items, and with each other, so the comparison is dependent and a two-sample test on Fisher-transformed correlations would be anticonservative. Table 9 reports the Williams test [24], which corrects for that dependence.

D What Survives a Change of Document

A metric meta-evaluation that never changes source document cannot distinguish legal sensitivity from familiarity with a form. We therefore add an out-of-domain split. FRJUDGE clauses come from identifiable source documents, which we recover from the raw export. Random splitting lets clauses from the same insurance form appear in both training and test, and insurance forms are highly templated. We compare the standard random split against a document-grouped split in which no source document straddles the boundary (Table 10); the difference estimates how much of the reported performance is form-specific. We re-run three of the seven configurations of Table 8 on that split: the scalar baseline and the two augmented mixtures. The remaining four differ from the baseline in what they are trained to predict rather than in what they can exploit about a source form, so the leakage they are exposed to is the same.

The corpus makes that harder than it sounds. Its 297 pairs come from only 21 source forms and the three largest hold 71% of them, so the obvious construction, sweeping the forms in random order and cutting at the cumulative 60/70% marks, lets a single large form overshoot a boundary and starve

Neural metric m	$r(m, h)$	$r(m, \ell)$	t	p
BERTScore-CamemBERTv2	0.344	0.349	5.05	<0.001
BERTScore-CamemBERTv2-Recall	0.396	0.443	4.43	<0.001
BERTScore-FlauBERT	0.484	0.389	2.61	0.010
LaBSE	0.424	0.362	3.66	<0.001
Sentence-CamemBERT	0.222	0.148	6.33	<0.001
Para-mUSE-mpnet	0.225	0.159	6.32	<0.001
mE5-base	0.318	0.316	5.40	<0.001
NLI-fwd (no hallucination)	0.320	0.388	5.66	<0.001
NLI-bwd (no omission)	0.307	0.258	5.36	<0.001
NLI-bidirectional (min)	0.365	0.350	4.68	<0.001

Table 9: Williams tests against the trivial length baseline $\ell = -||s| - |o||$, which reaches $r = 0.613$ with the human label h over all 297 items. $r(m, \ell)$ is each metric’s correlation with the length feature itself, the dependence the test corrects for. Positive t favours ℓ .

the split behind it. On seeds 42–46 that procedure returns test splits of 16, 87, 30, 0 and 88 pairs, one of them empty and one far too small to carry a correlation. We therefore choose the form-to-split assignment by a short local search over total deviation from the target sizes. Thousands of assignments hit 178/30/89 exactly, so each seed still selects a genuinely different partition while every split matches the random-split sizes of Table 8 pair for pair, and the two tables differ only in which forms are held out. One limitation is structural rather than incidental: the largest form is 37% of the corpus, larger than the 30% test quota, so it always falls in training and the grouped test set is always drawn from the remaining 188 pairs.

The comparison does not resolve. The scalar baseline’s correlation falls from 0.542 on the random split to 0.454 on the grouped one, a drop of 0.088 that is in the direction leakage would predict. It does not survive inspection. The per-seed differences are +0.169, −0.097, −0.416, −0.182 and +0.086: two of the five seeds *improve* under the harder split, and the grouped spread (± 0.143) is wider than the gap itself. The two augmented mixtures move by −0.002 and +0.026, which is to say not at all. As a control, the untrained NLI reference is fitted to nothing and should be indifferent to how the corpus is partitioned; it moves from 0.395 to 0.391, confirming that the two splits are otherwise comparable and that the variance above is the models’ and not the split’s.

We therefore do not claim that document-level leakage inflates the correlations of Section 6.6, and we hold our own evidence to the standard Section 6.3 applies elsewhere. What we can say is narrower and, we think, more useful to anyone planning a successor corpus: FRJUDGE is too small

and too concentrated to answer the question. Its grouped test split is always drawn from the 188 pairs the largest form leaves behind, at which size the seed-to-seed variation swamps the effect being measured. A corpus intended to support an out-of-domain claim needs its documents spread far more evenly than three forms holding 71% of the data, and this is the aspect of the design we would change first.

E Stratifying by Length Change

We further break the test predictions down by the magnitude of the length change between original and simplification. If a metric’s correlation is concentrated in the buckets where the simplification is much shorter, it is detecting deletion rather than legal change, the hypothesis raised in Section 6.3. Table 11 reports Pearson r inside each tercile of $||s| - |o||$. Within a tercile the length signal is largely held constant, so a metric that retains its correlation there is using something other than deletion; one whose correlation collapses was reading length all along. This is a weaker form of the same dissociation logic as Section 3: it holds a confound approximately fixed by conditioning rather than exactly fixed by construction.

The result qualifies Section 6.3 in a way that is worth stating carefully, because it cuts against our own argument. The length feature’s correlation is overwhelmingly a *between*-tercile effect: pooled over the corpus it reaches 0.613, but inside the small and medium terciles it falls to 0.243 and 0.165. Once items of comparable length change are compared with each other, knowing the length change tells you little. It recovers to 0.585 only in the tercile where the simplification is much shorter than the original, which is exactly where deletion

Model	r	ρ	RMSE	over%	ident.	unrel.	LEGALEDIT	
							pass	margin
JudgeBERT-Scalar (baseline)	0.454 $\pm$ 0.14	0.282	2.49	24.0	0	0	0	–
JudgeBERT-DA	0.190 $\pm$ 0.21	0.140	2.63	33.0	0	100	9	0.004
JudgeBERT-DA+LegalEdit	0.149 $\pm$ 0.13	0.319	2.65	28.1	0	49	0	0.030
<i>NLI-bidirectional</i> (untrained)	0.391	0.401	2.20	45.7	1	–	–	0.670
Human ceiling	0.597	0.620	3.27	–	–	–	–	–

Table 10: Document-grouped split: no source form appears in both training and test. Columns are as in Table 8, and the splits are size-matched to it (178/30/89). The untrained NLI reference is recomputed on the grouped splits rather than carried over, since a corpus-level correlation is not comparable to a document-grouped one; the human ceiling is a property of the annotations and does not depend on the split.

Metric	r within tercile			all
	small	medium	large	
$- s - o $	0.243	0.165	0.585	0.613
BERTScore-FlauBERT	0.198	0.449	0.413	0.484
LaBSE	0.174	0.532	0.240	0.424
NLI-bidirectional	0.058	0.269	0.291	0.365
Surface-Ridge	0.118	0.305	0.540	0.620
JudgeBERT-Scalar	0.042	0.149	0.602	0.496

Table 11: Correlation with the human label *within* terciles of the absolute length change, pooling test predictions over seeds. Restricting to a tercile removes most of the length variance; a metric that measures legal meaning should degrade far less than one that measures deletion.

is available to be detected. The neural baselines have the opposite profile: in the medium tercile, where the length feature retains almost nothing, BERTScore over FlauBERT holds 0.449 against a pooled 0.484 and LaBSE rises to 0.532 against a pooled 0.424. The supervised surface ridge tracks the raw length feature bucket for bucket, which is what one would expect if it is largely a length model with extra terms. The small tercile is hard for everything, which we read as a floor effect rather than as a result: when the two sentences are close to the same length, none of these signals has much to go on.

The honest reading is therefore narrower than “legal meaning preservation is length preservation”. The corpus-level ranking is dominated by length, and any comparison that omits a length control will attribute to a metric what it should attribute to that confound. But the neural metrics are not *only* reading length. Where the length signal is weakest, they are the ones that hold up, keeping or exceeding the correlation they had over the whole corpus. Whatever they are reading in that stratum, it is not the length change. What Section 5 adds is

that the residual signal they retain is still not legal sensitivity, since the same metrics move almost not at all when the law is changed and the length is held exactly fixed. The two analyses answer different questions, and a metric has to survive both.

F Reproducibility

Every number in this paper is produced by a single pipeline that runs end to end on a laptop; the largest model involved is CamemBERTv2-base (112M parameters) and no GPU cluster is required. We release the experiment code, the reconstruction of FRJUDGE from the raw export (which recovers annotator identity and source document, both dropped from the public JSONL), the document-grouped splits, the LEGALEDIT generator and scoring harness, and the result files behind every table.

One caveat on exact reproduction: our training runs use Apple MPS kernels, which are not bit-wise deterministic, so a fixed seed reproduces a run only approximately: repeat runs of the scalar configuration at seed 42 differ in test-set r by several hundredths, which is comparable to the spread across seeds. We therefore report means and standard deviations over five seeds and treat any difference smaller than that spread as unresolved, the same discipline Section 6.3 asks of everyone else, applied to ourselves. Anyone reproducing this on CUDA should expect tighter agreement and correspondingly narrower error bars.

Two further details matter for anyone reproducing the diagnostic. The first is what the probes share with training, which differs by regime and which we state precisely because conflating the two is one of the failure modes the protocol is meant to prevent. The *identical* probe is built from statutory sentences (the Quebec Automobile Insurance Act and the Highway Safety Code) while the identical

pairs in the -DA training mixture are built from FRJUDGE’s insurance-form clauses, so a model passes that probe only by generalising across source documents. The LEGAEDIT probe is likewise generated from statutory sentences, while the minimally-edited negatives added by the DA+LEGAEDIT variant are generated from FRJUDGE originals: same perturbation rules, disjoint sentences. The *unrelated* probe is the weakest of the three in this respect: both it and the unrelated pairs in the training mixture are drawn from the same two statutes, under different random seeds, so a high unrelated pass rate for a -DA model should be read as in-distribution and we do not treat it as evidence of generalisation. Second, the ordinal Krippendorff coefficient we report is computed with the same library and the same reliability matrix as the nominal one, so the difference between 0.104 and 0.325 is attributable to the distance function alone.